MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, MICROPROCESSOR COMPATIBLE 8-BIT DIGITAL-TO-ANALOG CONVERTER, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, range-programmable, output voltage digital-to-analog converter. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Logic coding	Technology	Output voltage range
01	Binary	Monolithic	0 V to +2.56 V 0 V to +10.0 V

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package

1.3 Absolute maximum ratings.

Positive supply voltage (V_{CC}) to digital return	+18 V dc
Digital input voltage to digital return	0 V to +7.0 V
Output short circuit duration (to ground)	Indefinite
Output short circuit duration (to V_{CC})	Momentary
Power dissipation	450 mW
Lead temperature (soldering, 10 seconds)	+300°C
Storage temperature	-65°C to +150°C
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Positive supply voltage range (V_{CC}):	
2.56 V output range	4.5 V to 5.5 V
10.0 V output range	11.4 V to 16.5 V
Ambient operating temperature range (T_A)	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{J-C}	Maximum θ_{J-A}
16-lead dual-in-line	E	450 mW at T_A = +125°C	30°C/W	100°C/W

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Functional circuit diagram and terminal connections. The functional circuit diagram and terminal connections shall be as shown on figure 1.

3.2.2 Truth table. The truth table shall be as shown on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified. The following electrical test conditions shall apply to all parameters unless otherwise specified in table I:

V_{CC} = 15.0 V
$\overline{CS}$ = logic "0"
$\overline{CE}$ = logic "0"
$-55^{\circ}C < T_A < +125^{\circ}C$
Load resistance R_L = 10 kΩ

V_{FSR} = 10 V
Logic "0" = 0.8 V
Logic "1" = 2.0 V

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol 1/	Conditions (see 3.4 and figure 3) unless otherwise specified 2/	Limits		Units
			Min	Max	
Resolution		All device types	+8		Bits
Supply current from V_{CC}	I_{CC}	Input bits = 1111 1111 V_{CC} = 15.0 V	+1	+25	mA
Input low current	I_{IL}	V_{IN}(LOGIC) = 0 V V_{CC} = 15.0 V (Each input measured separately)	-100	+100	µA
Input high current	I_{IH}	V_{IN}(LOGIC) = 2.4 V (Each input measured separately)	-100	+100	µA
		V_{IN}(LOGIC) = 5.5 V (Each input measured separately)	-100	+100	µA
Output short circuit current	I_{OSC}	Input bis = 1111 1111 V_{SFR} = 10 V T_A = +25°C		+30	mA
		Input bits = 1111 1111 V_{SFR} = 2.56 V T_A = +25°C		+30	mA
Zero voltage error 3/	V_{ZE}	Input bits = 0000 0000 V_{FSR} = 10 V (Initial) T_A = +25°C -55°C ≤ T_A ≤ +125°C (End point) T_A = +25°C	 -0.5 -1.0 -0.6	 +0.5 +1.0 +0.6	 LSB LSB LSB
		Input bits = 0000 0000 V_{FSR} = 2.56 V (Initial) T_A = +25°C -55°C ≤ T_A ≤ +125°C (End point) T_A = +25°C	 -0.5 -1.0 -0.6	 +0.5 +1.0 +0.6	 LSB LSB LSB

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol 1/	Conditions (see 3.4 and figure 3) unless otherwise specified 2/	Limits Min	Limits Max	Units
Full scale voltage error	V_{FS}	Input bits = 1111 1111 V_{FSR} = 10 V (Initial) T_A = +25°C -55°C ≤ T_A ≤ +125°C (End point) T_A = +25°C	 -1.00 -1.75 -1.75	 +1.00 +1.75 +1.75	 LSB LSB LSB
		Input bits = 1111 1111 V_{FSR} = 2.56 V (Initial) T_A = +25°C -55°C ≤ T_A ≤ +125°C (End point) T_A = +25°C	 -1.00 -1.75 -1.75	 +1.00 +1.75 +1.75	 LSB LSB LSB
Power supply sensitivity from V_{CC} at full scale	$+P_{SS1}$	Input bits = 1111 1111 V_{FSR} = 10 V 11.4 V ≤ V_{CC} ≤ 14.0 V 14.0 ≤ V_{CC} ≤ 16.5 V	 -.03 -.03	 +.03 +.03	 LSB/%PS LSB/%PS
	$+P_{SS2}$	Input bits = 1111 1111 V_{FSR} = 2.56 V 4.5 V ≤ V_{CC} ≤ 5.0 V 5.0 ≤ V_{CC} ≤ 5.5 V	 -.03 -.03	 +.03 +.03	 LS µP. LSB/%PS
Integral linearity error 4/	L_E	All codes test V_{FSR} = 10 V (Initial) T_A = +25°C -55°C ≤ T_A ≤ +125°C (End point) T_A = +25°C	 -.25 -.375 -.375	 +.25 +.375 +.375	 LSB LSB LSB
		All codes test V_{FSR} = 2.56 V (Initial) T_A = +25°C -55°C ≤ T_A ≤ +125°C (End point) T_A = +25°C	 -.25 -.375 -.375	 +.25 +.375 +.375	 LSB LSB LSB
Differential linearity error	D_{LE}	Verified by all-codes integral linearity error test	-1.0	+1.0	LSB
Data hold time	t_{DH}	V_{FSR} = 10 V Strobe pulse width = 200 ns (See figures 3 and 4)	10		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol 1/	Conditions (see 3.4 and figure 3) unless otherwise specified 2/	Limits		Units
			Min	Max	
Data setup time	t_{DS}	V_{FSR} = 10 V T_A = +25°C Strobe pulse width = 200 ns	200		ns
		T_A = +125°C, -55°C Strobe pulse width = 300 ns (See figures 3 and 4)	300		ns
Settling time	t_{SLH1}	V_{FSR} = 10 V R1 = 10 kΩ Input bits = 0000 0000 to 1111 1111 0 V to 9.96 V ±1/2 LSB (See figures 4 and 5)		3.0	µs
Settling time	t_{SHL1}	V_{FSR} = 10 V R1 = 10 kΩ Input bits = 1111 1111 to 0000 0000 9.96 V to 0 V ±1/2 LSB (See figures 4 and 5)		3.0	µs
Settling time	t_{SLH2}	V_{FSR} = 2.56 V R1 = 2 kΩ Input bits = 0000 0000 to 1111 1111 0 V to 2.55 V ±1/2 LSB (See figures 4 and 5)		1.5	µs
Settling time	t_{SHL2}	V_{FSR} = 2.56 V R1 = 2 kΩ Input bits = 1111 1111 to 0000 0000 2.55 V to 0 V ±1/2 LSB (See figures 4 and 5)		1.5	µs
Output noise voltage	NO	All inputs = 1111 1111 10 Hz ≤ BW ≤ 100 kHz T_A = +25°C (See figure 6)		250	µV rms

1/ See 6.4 for symbols and definitions.
2/ Unless otherwise specified, all tests are performed in the 10 V range. The scale factor is V_{FSR}/256 LSB (i.e., for V_{FSR} = 10 V, the scale factor is 10 V/256 LSB = 39.1 mV/LSB.)
3/ Zero voltage error is the deviation from zero volts at the output with no external load.
4/ The offset error used to calculate the ideal integral linearity is measured by applying a -0.5 mA load to the device.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 56 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004) 1/	1, 2, 3, 4, 5, 6	1, 2, 3, 4, 5, 6
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6, 9, 12	1, 2, 3, 4, 5, 6, 9, 12
Group C end-point and group B, class S, electrical parameters (method 5005)	1, 2, 3, 4, 5, 6 and table IV delta limits	1, 4, and table IV delta limits
Group D end-point electrical parameters (method 5005) 2/	1, 2, 3, 4, 5, 6	1, 4

1/ PDA applies to subgroup 1 (see 4.2c).
2/ See 4.4.4.

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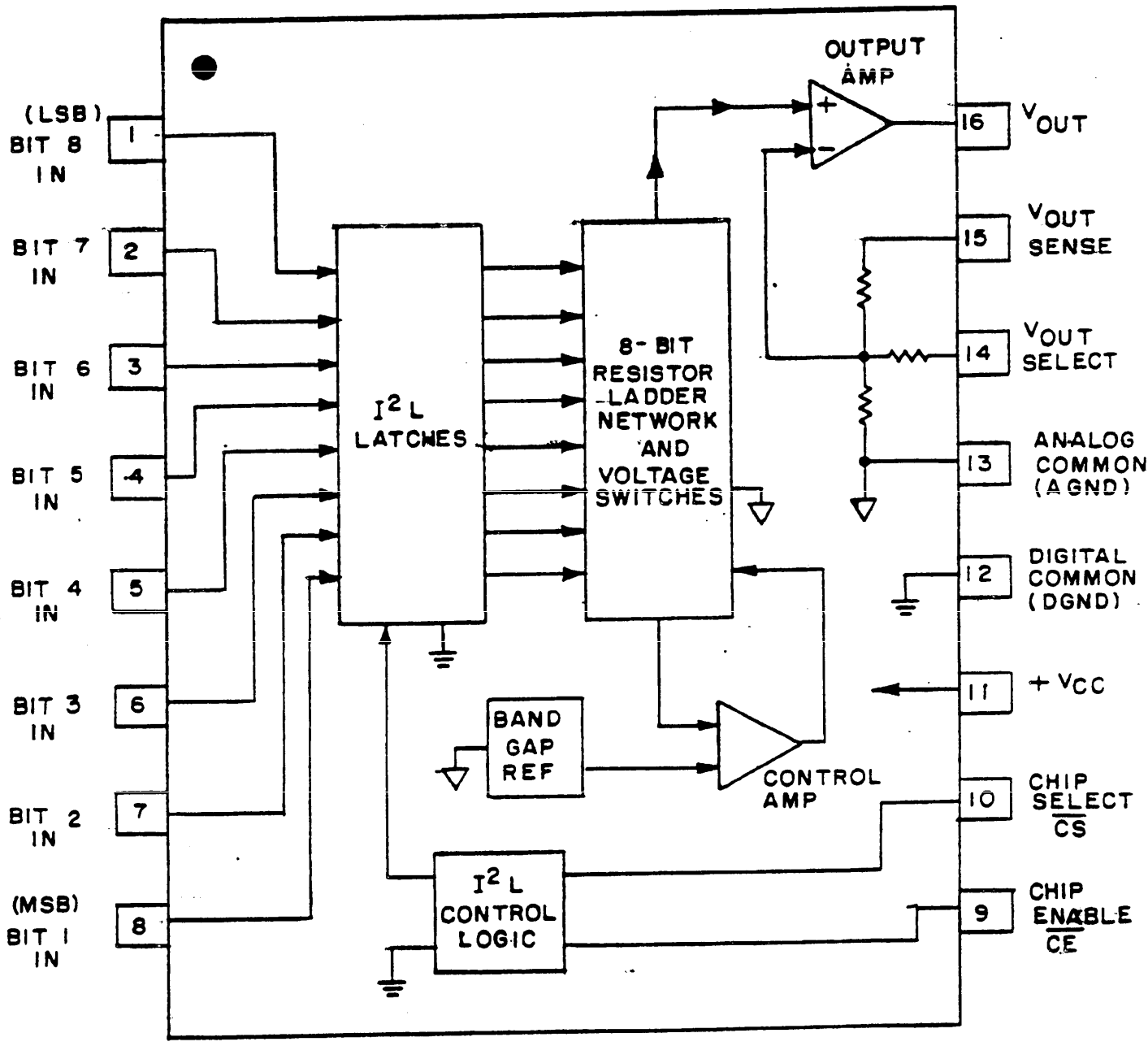


FIGURE 1. Functional schematic and terminal connections.

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Input-output relationship

Digital input code			Output voltage	
Binary	Hexadecimal	Decimal	2.56 V range	10.00 V range
0000 0000	00	0	0	0
0000 0001	01	1	0.010 V	0.039 V
0000 0010	02	2	0.020 V	0.078 V
0000 1111	0F	15	0.150 V	0.586 V
0001 0000	10	16	0.160 V	0.625 V
0111 1111	7F	127	1.270 V	4.961 V
1000 0000	80	128	1.280 V	5.000 V
1100 0000	C0	192	1.920 V	7.500 V
1111 1111	FF	255	2.55 V	9.961 V

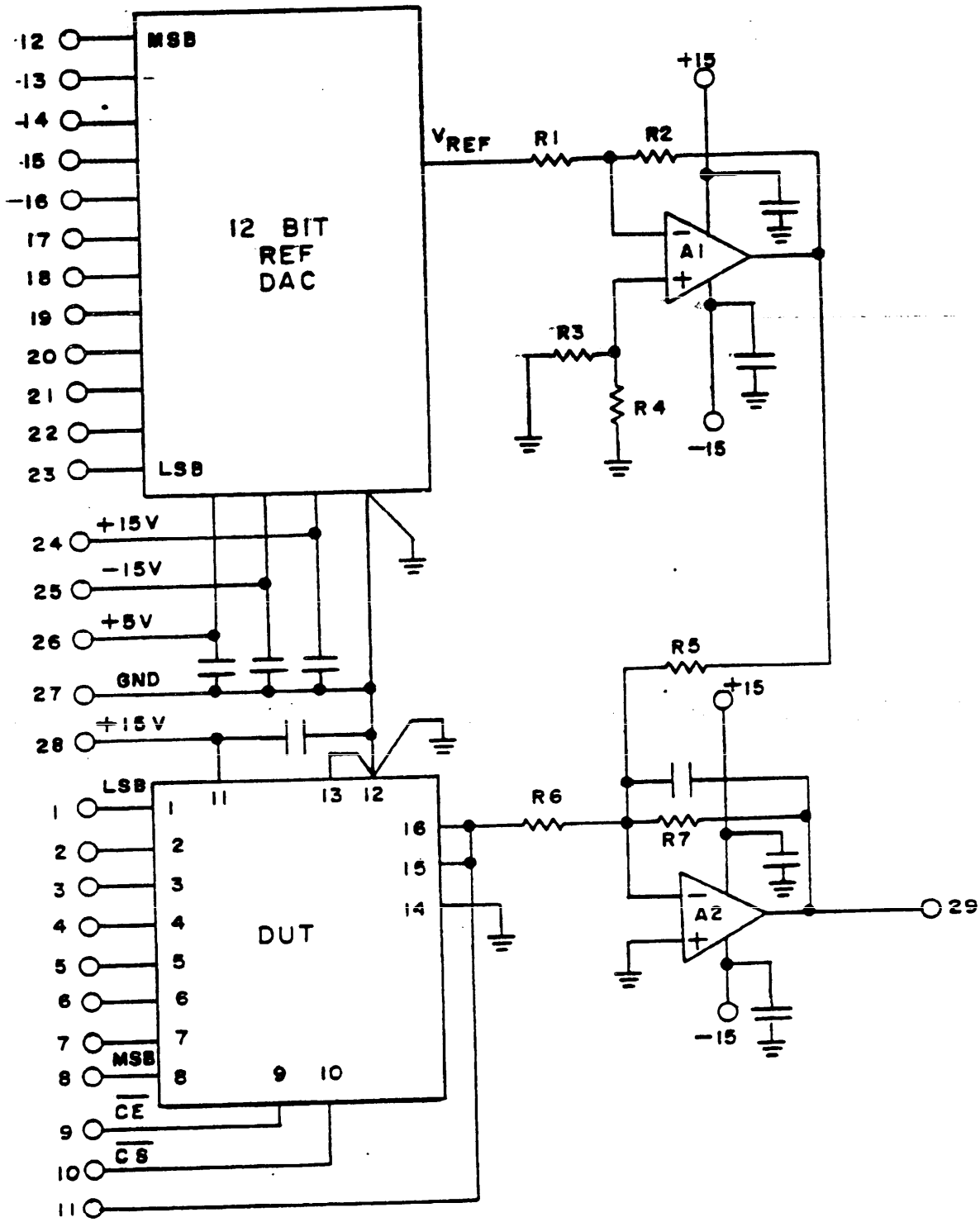
Control logic truth table

Input data	$\overline{CE}$	$\overline{CS}$	DAC data	Latch condition
0	0	0	0	"transparent"
1	0	0	1	"transparent"
0	⌐↑	0	0	latching
1	⌐↑	0	1	latching
0	0	⌐↑	0	latching
1	0	⌐↑	1	latching
X	1	X	previous data	latched
X	X	1	previous data	latched

Notes: X = Does not matter
⌐↑ = Logic threshold at positive-going transition

FIGURE 2. Input-output relationship and truth table.

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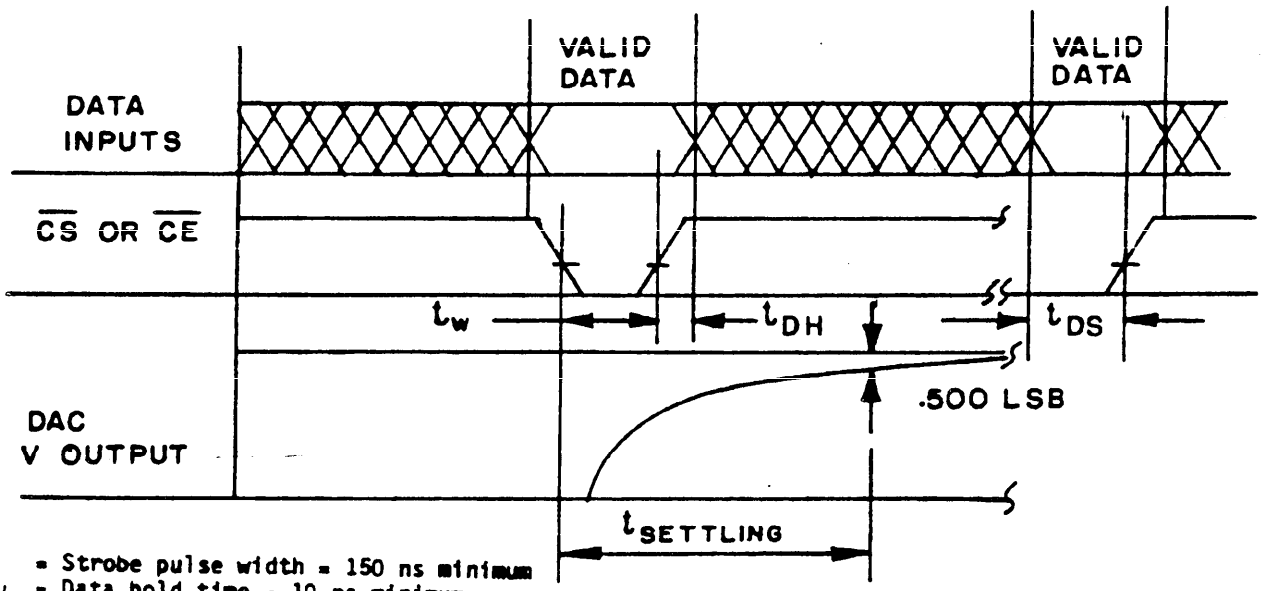


NOTES:
1. All bypass capacitors are 0.1 µF ceramic.
2. R1 - R4 = 5.11 kΩ 1 percent.
   R5 - R6 = 5.11 kΩ .01 percent.
   - R7 = 511 kΩ .01 percent
   A1 - A2 = OP07 or equivalent.
3. See figure 2 truth table.

FIGURE 3. Test circuit for static tests.

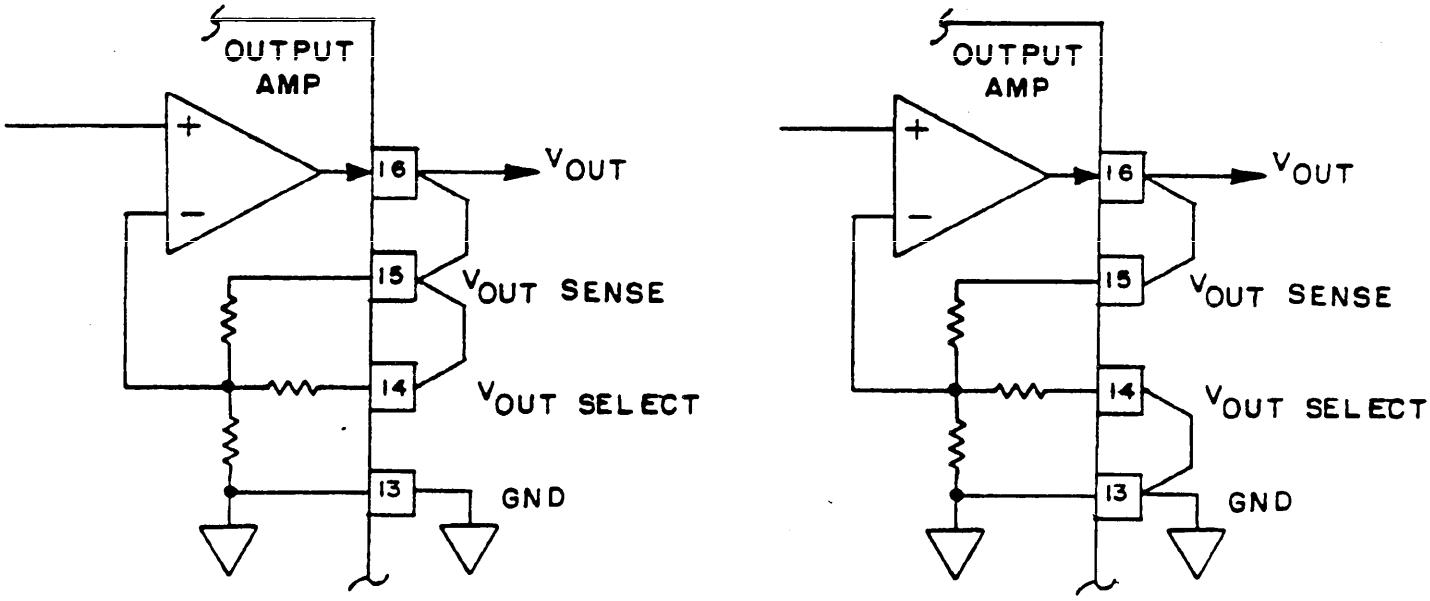
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TIMING DIAGRAM



t_w = Strobe pulse width = 150 ns minimum
t_{DH} = Data hold time = 10 ns minimum
t_{DS} = Data setup time = 200 ns minimum
$t_{settling}$ = DAC output settling time to ±1/2 LSB

OUTPUT VOLTAGE RANGE CONNECTIONS

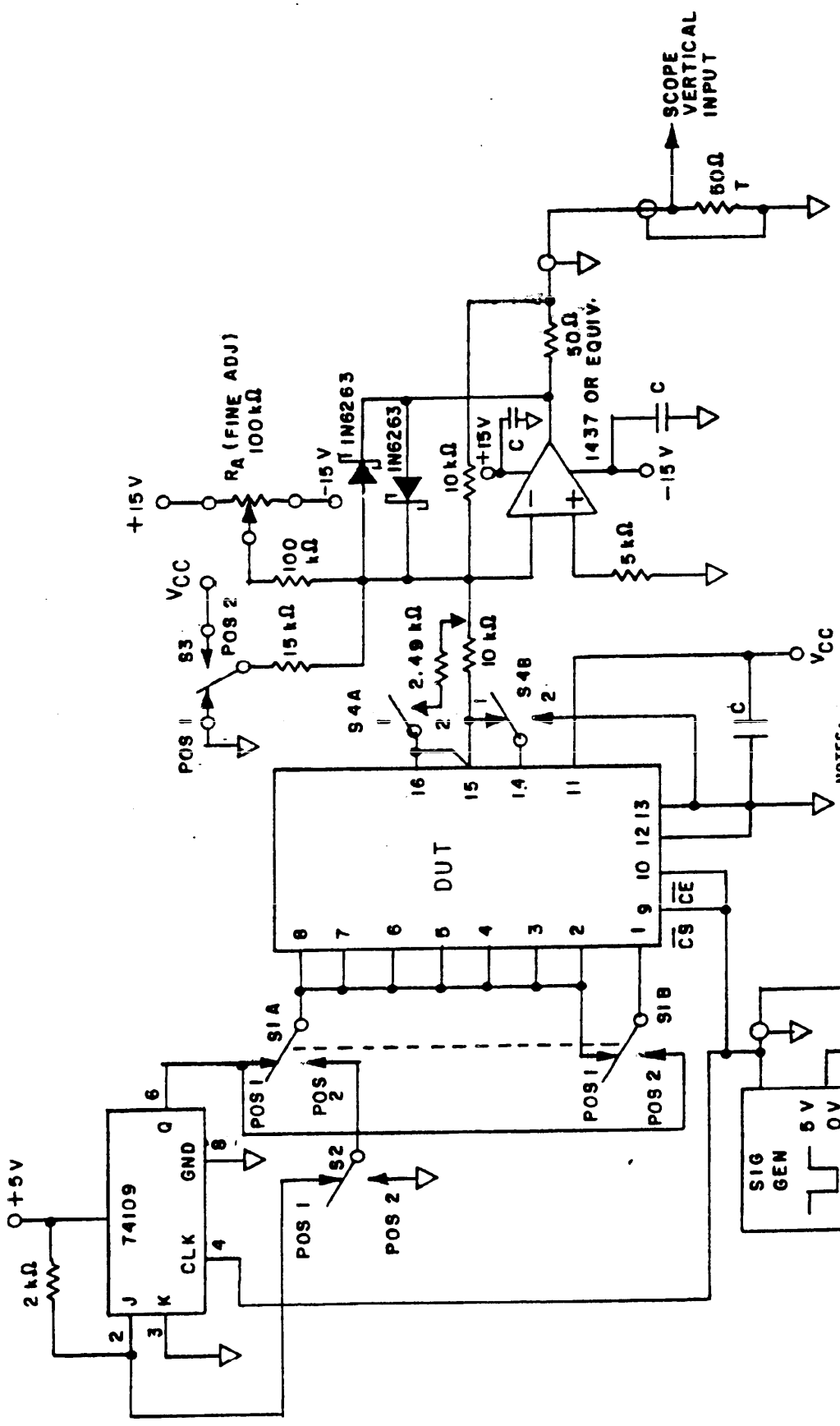


a. 0 V to 2.56 V output range

b. 0 V to 10 V output range

FIGURE 4. Timing diagram and output voltage range connections.

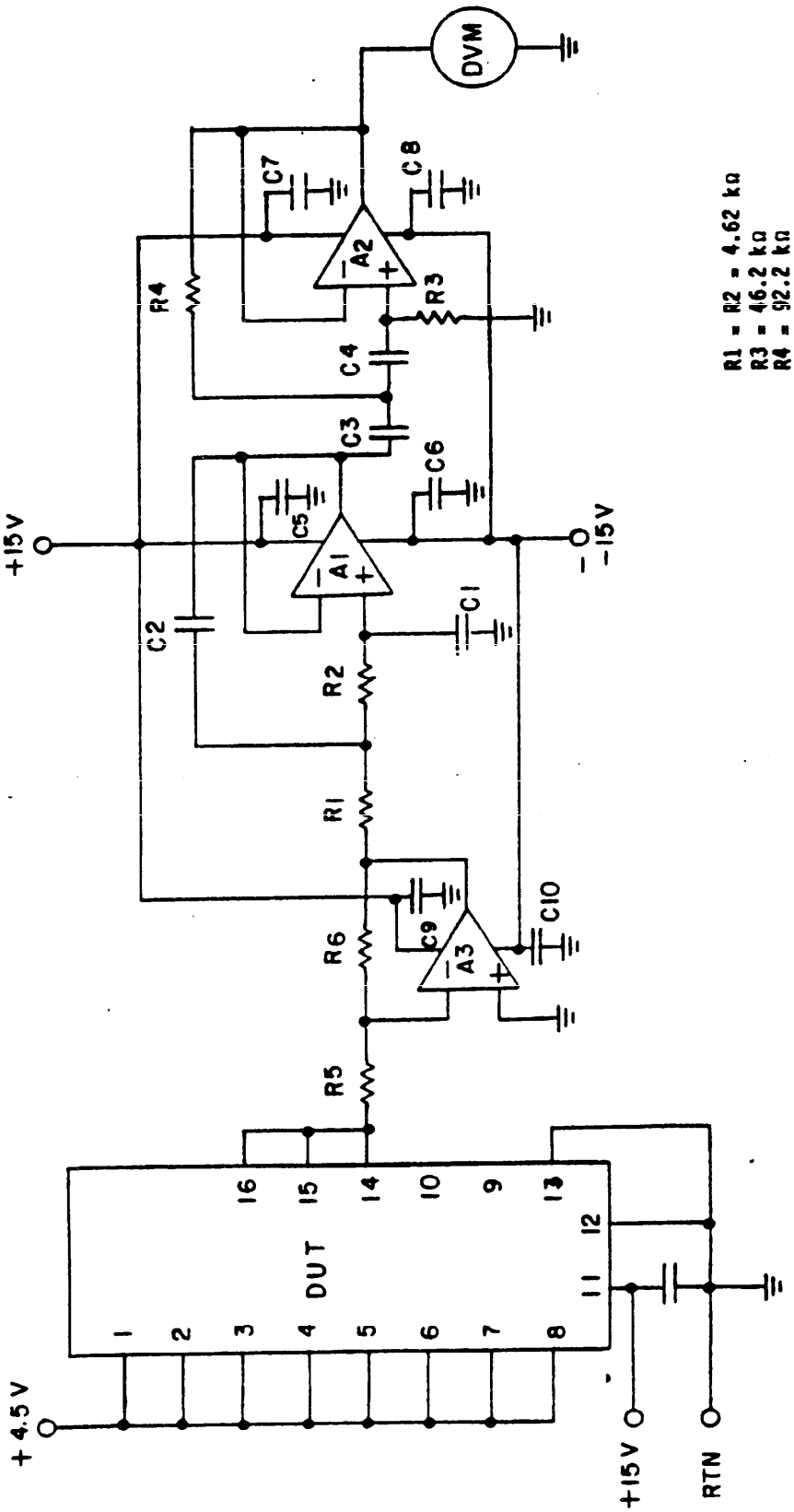
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NOTES:
1. C = .47 µF tantalum capacitor in parallel with a 0.1 µF ceramic capacitor.
2. Calibration: Set S1 in position 2, set S2 in position 2 and set S3 in position 1. Adjust potentiometer, R_A, until a 1 kHz square wave appears on the oscilloscope. (Vertical gain = 20 mV/cm) Adjust the scope vertical calibration control so that the square wave peak-to-peak voltage is 2 cm high. (1 LSB = 40 mV/cm)
3. Test: Adjust the scope horizontal timing and measure the high-to-low and low-to-high settling times, 1/2 LSB from the final value. Set S1 in position 1 and measure the high-to-low settling times 1/2 LSB from the final value (as seen on the oscilloscope). Repeat the calibration with S2 in position 2 and S3 in position 1. Set S1 in position 1 and measure the low-to-high settling times 1/2 LSB from the final value (as seen on the oscilloscope).

FIGURE 5. Test circuit for settling time.

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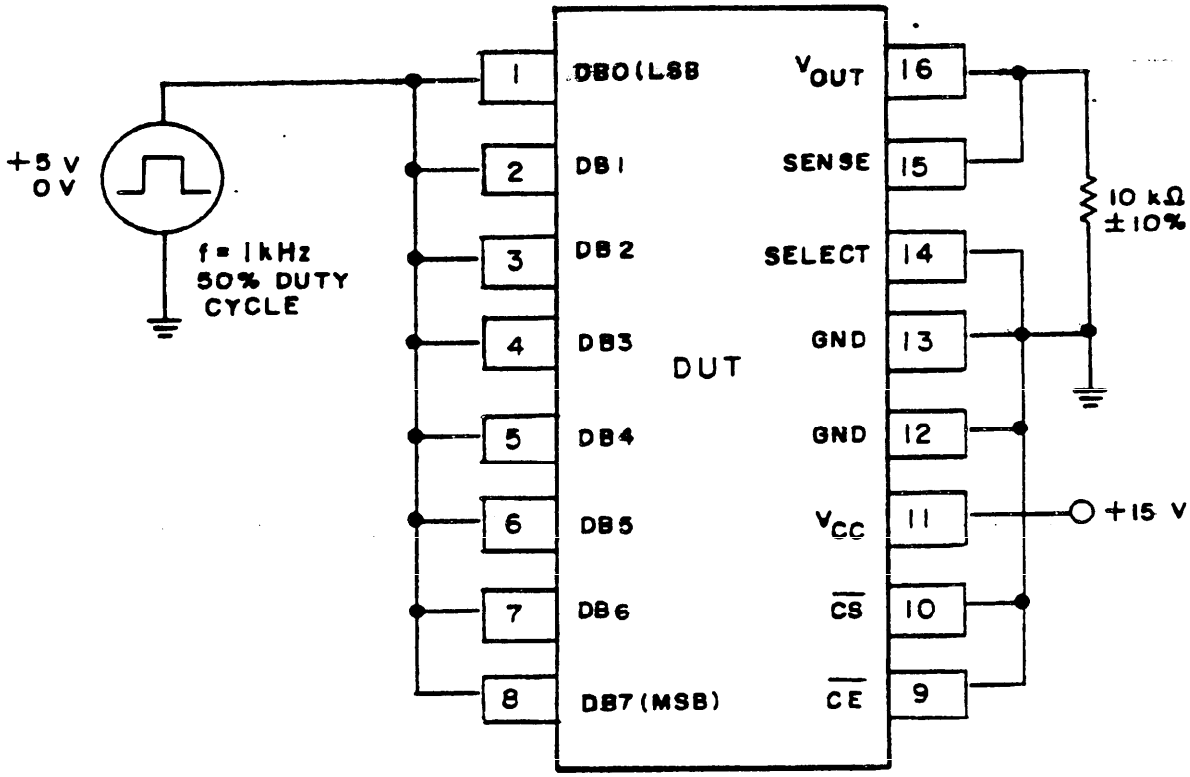


R1 = R2 = 4.62 kΩ
R3 = 46.2 kΩ
R4 = 92.2 kΩ
R5 = 10.0 kΩ
R6 = 100 kΩ
C1 = 270 pF.
C2 = 560 pF.
C3 = C4 = .27 µF.
C5 through C10 see note 1

NOTES:
1. Unless otherwise stated, all bypass capacitors are 4.7 µF tantalum parallel with 0.1 µF ceramic.
2. The meter, DVM, shall read rms and have a bandwidth greater than 100 kHz.
3. A1-A3 shall have a gain-bandwidth product greater than 1 MHz and input noise density less than 10 nV/square root hertz.
4. All ground connections shall return to a single unipoint ground at DUT pin 12.

FIGURE 6. Test circuit for output e tests.

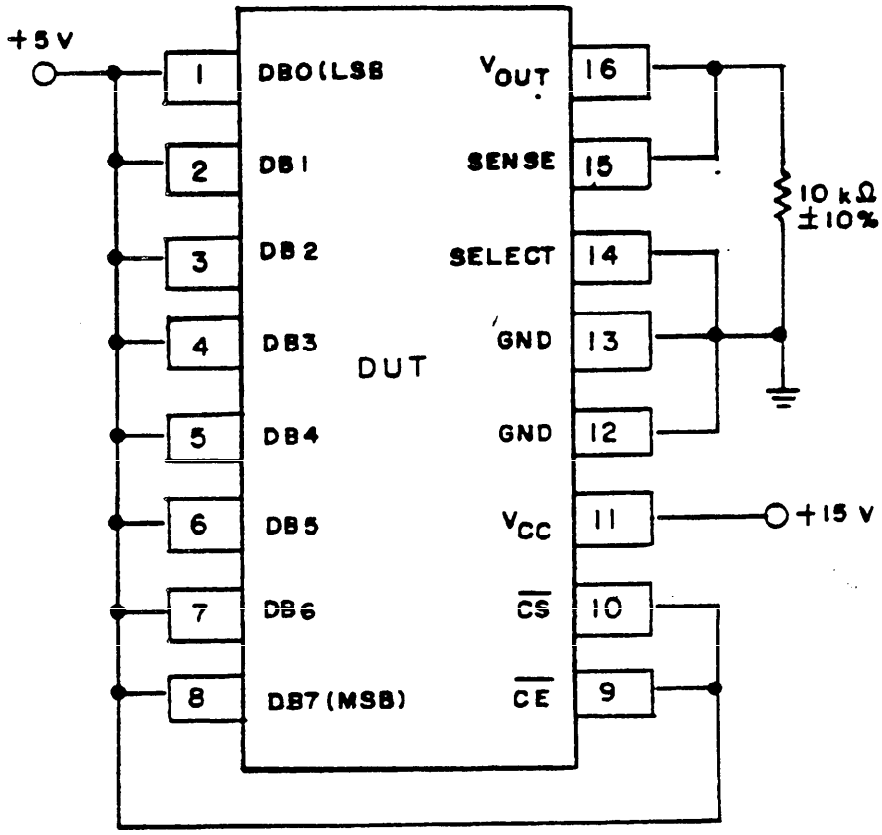
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NOTE: Bypass supply with a 1 µF tantalum capacitor and a 0.1 µF ceramic capacitor to common.

FIGURE 7. Dynamic test circuit, burn-in and steady-state life test.

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NOTE: Bypass supply with a 1 µF tantalum capacitor and a 0.1 µF ceramic capacitor to common.

FIGURE 8. Static test circuit, burn-in and steady-state life test.

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TABLE III. Group A inspection.

Subgroup	Symbol	Test no.	Applied voltage (Fig 3)	Applied DUT digital code-word	Applied reference DAC digital code-word	Measurement sense lines reference pin no. 25			Equations	Limits		Unit
						Pin	Value	Unit		Min	Max	
			28 V_{CC}(V)	8(-----)1 MSB(-)LSB	12(---------)23 MSB(-------)LSB							
1 T_A = +25°C	I_{CC}	001	+15.0	1111 1111	1111 1111 0000	28	I001	mA	I_{CC} = I001	1	25	mA
	I_{IL1}	002	"	0000 0000	0000 0000 0000	1	I002	µA	I_{IL1} = I002	-100	+100	µA
	I_{IL2}	003	"	" "	" " "	2	I003	"	I_{IL2} = I003	"	"	"
	I_{IL3}	004	"	" "	" " "	3	I004	"	I_{IL3} = I004	"	"	"
	I_{IL4}	005	"	" "	" " "	4	I005	"	I_{IL4} = I005	"	"	"
	I_{IL5}	006	"	" "	" " "	5	I006	"	I_{IL5} = I006	"	"	"
	I_{IL6}	007	"	" "	" " "	6	I007	"	I_{IL6} = I007	"	"	"
	I_{IL7}	008	"	" "	" " "	7	I008	"	I_{IL7} = I008	"	"	"
	I_{IL8}	009	"	" "	" " "	8	I009	"	I_{IL8} = I009	"	"	"
	I_{IH1}	010	"	1111 1111	1111 1111 0000	1	I010	"	I_{IH1} = I010 V_{IN}(LOGIC) = 2.4 V	"	"	"
	I_{IH2}	011	"	" "	" " "	2	I011	"	I_{IH2} = I011	"	"	"
	I_{IH3}	012	"	" "	" " "	3	I012	"	I_{IH3} = I012	"	"	"
	I_{IH4}	013	"	" "	" " "	4	I013	"	I_{IH4} = I013	"	"	"
	I_{IH5}	014	"	" "	" " "	5	I014	"	I_{IH5} = I014	"	"	"
	I_{IH6}	015	"	" "	" " "	6	I015	"	I_{IH6} = I015	"	"	"
	I_{IH7}	016	"	" "	" " "	7	I016	"	I_{IH7} = I016	"	"	"
	I_{IH8}	017	"	" "	" " "	8	I017	"	I_{IH8} = I017	"	"	"
	I_{IH9}	018	"	" "	" " "	1	I018	"	I_{IH9} = I018 V_{IN}(LOGIC) = 5.5 V	"	"	"
	I_{IH10}	019	"	" "	" " "	2	I019	"	I_{IH10} = I019	"	"	"
	I_{IH11}	020	"	" "	" " "	3	I020	"	I_{IH11} = I020	"	"	"
	I_{IH12}	021	"	" "	" " "	4	I021	"	I_{IH12} = I021	"	"	"
	I_{IH13}	022	"	" "	" " "	5	I022	"	I_{IH13} = I022	"	"	"
	I_{IH14}	023	"	" "	" " "	6	I023	"	I_{IH14} = I023	"	"	"
	I_{IH15}	024	"	" "	" " "	7	I024	"	I_{IH15} = I024	"	"	"
	I_{IH15}	025	"	" "	" " "	8	I025	"	I_{IH16} = I025	"	"	"
	I_{OSC}	026	"	" "	" " "	11	I026	mA	I_{OSC} = I026 V_{FSR} = 10.0 V		+30	mA
		027	+5.0	" "	" " "	11	I027	mA	I_{OSC} = I027 V_{FSR} = 2.56 V		+30	mA
	V_{ZE}	028	+15.0	0000 0000	0000 0000 0000	29	E001	V	V_{ZE1} = -.256*(E001) V_{FSR} = 10.0 V	-0.5	+0.5	LSB
		029	+5.0	0000 0000	0000 0000 0000	"	E002	"	V_{ZE2} = -1.00*(E002) V_{FSR} = 2.56 V	-0.5	+0.5	"
	V_{FS}	030	+15.0	1111 1111	1111 1111 0000	"	E003	"	V_{FS1} = -.256*(E003) V_{FSR} = 10.0 V	-1.0	+1.0	"
		031	+5.0	" "	" " "	"	E004	"	V_{FS2} = -1.00*(E004) V_{FSR} = 2.56 V	-1.0	+1.0	"
	$+P_{SS1}$	032	+16.5	" "	" " "	"	E005	"				
		033	+14.0	" "	" " "	"	E006	"	$+P_{SS1a}$ = .014*(E005-E006)	-.03	+.03	LSB/%PS
		034	+11.4	" "	" " "	"	E007	"	$+P_{SS1b}$ = .014*(E006-E007)	"	"	"
	$+P_{SS2}$	035	+5.5	" "	" " "	"	E008	"				
		036	+5.0	" "	" " "	"	E009	"	$+P_{SS2a}$ = .10*(E008-E009)	"	"	"
		037	+4.5	" "	" " "	"	E010	"	$+P_{SS2b}$ = .10*(E009-E010)	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltage (Fig 3)	Applied DUT digital code-word	Applied reference DAC digital code-word	Measurement sense lines reference pin no. 25			Equations	Limits		Unit
			28 V_{CC}(V)	8(-----)1 MSB(-)LSB	12(---------)23 MSB(-------)LSB	Pin	Value	Unit		Min	Max	
2 T_A = +125°C	I_{CC}	038	+15.0	1111 1111	1111 1111 0000	28	I028	mA	I_{CC} = I028	+1	+25	mA
	I_{IL1}	039	"	0000 0000	0000 0000 0000	1	I029	µA	I_{IL1} = I029	-100	+100	µA
	I_{IL2}	040	"	" "	" " "	2	I030	"	I_{IL2} = I030	"	"	"
	I_{IL3}	041	"	" "	" " "	3	I031	"	I_{IL3} = I031	"	"	"
	I_{IL4}	042	"	" "	" " "	4	I032	"	I_{IL4} = I032	"	"	"
	I_{IL5}	043	"	" "	" " "	5	I033	"	I_{IL5} = I033	"	"	"
	I_{IL6}	044	"	" "	" " "	6	I034	"	I_{IL6} = I034	"	"	"
	I_{IL7}	045	"	" "	" " "	7	I035	"	I_{IL7} = I035	"	"	"
	I_{IL8}	046	"	" "	" " "	8	I036	"	I_{IL8} = I036	"	"	"
	I_{IH1}	047	"	1111 1111	1111 1111 0000	1	I037	"	I_{IH1} = I037 V_{IN}(LOGIC) = 2.4 V	"	"	"
	I_{IH2}	048	"	" "	" " "	2	I038	"	I_{IH2} = I038	"	"	"
	I_{IH3}	049	"	" "	" " "	3	I039	"	I_{IH3} = I039	"	"	"
	I_{IH4}	050	"	" "	" " "	4	I040	"	I_{IH4} = I040	"	"	"
	I_{IH5}	051	"	" "	" " "	5	I041	"	I_{IH5} = I041	"	"	"
	I_{IH6}	052	"	" "	" " "	6	I042	"	I_{IH6} = I042	"	"	"
	I_{IH7}	053	"	" "	" " "	7	I043	"	I_{IH7} = I043	"	"	"
	I_{IH8}	054	"	" "	" " "	8	I044	"	I_{IH8} = I044	"	"	"
	I_{IH9}	055	"	" "	" " "	1	I045	"	I_{IH13} = I045 V_{IN}(LOGIC) = 5.5 V	"	"	"
	I_{IH10}	056	"	" "	" " "	2	I046	"	I_{IH14} = I046	"	"	"
	I_{IH11}	057	"	" "	" " "	3	I047	"	I_{IH15} = I047	"	"	"
	I_{IH12}	058	"	" "	" " "	4	I048	"	I_{IH16} = I048	"	"	"
	I_{IH13}	059	"	" "	" " "	5	I049	"	I_{IH17} = I049	"	"	"
	I_{IH14}	060	"	" "	" " "	6	I050	"	I_{IH18} = I050	"	"	"
	I_{IH15}	061	"	" "	" " "	7	I051	"	I_{IH19} = I051	"	"	"
	I_{IH16}	062	"	" "	" " "	8	I052	"	I_{IH20} = I052	"	"	"
	I_{OSC}	063	"	" "	" " "	11	I053	mA	I_{OSC} = I053 V_{FSR} = 10.0 V		+30	mA
		064	+5.0	" "	" " "	11	I054	mA	I_{OSC} = I054 V_{FSR} = 2.56 V		+30	mA
	V_{ZE}	065	+15.0	0000 0000	0000 0000 0000	29	E011	V	V_{ZE1} = -.256*(E011) V_{FSR} = 10.0 V	-1.0	+1.0	LSB
		066	+5.0	0000 0000	0000 0000 0000	"	E012	"	V_{ZE2} = -1.00*(E012) V_{FSR} = 2.56 V	-1.0	+1.0	"
	V_{FS}	067	+15.0	1111 1111	1111 1111 0000	"	E013	"	V_{FS1} = -.256*(E013) V_{FSR} = 10.0 V	-1.75	+1.75	"
		068	+5.0	" "	" " "	"	E014	"	V_{FS2} = -1.00*(E014) V_{FSR} = 2.56 V	-1.75	+1.75	"
	$+P_{SS1}$	069	+16.5	" "	" " "	"	E015	"				
		070	+14.0	" "	" " "	"	E016	"	$+P_{SS1a}$ = .014*(E015-E016)	-.03	+.03	LSB/%PS
		071	+11.4	" "	" " "	"	E017	"	$+P_{SS1b}$ = .014*(E016-E017)	"	"	"
	$+P_{SS2}$	072	+5.5	" "	" " "	"	E018	"				
		073	+5.0	" "	" " "	"	E019	"	$+P_{SS2a}$ = .10*(E018-E019)	"	"	"
		074	+4.5	" "	" " "	"	E020	"	$+P_{SS2b}$ = .10*(E019-E020)	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltage (Fig 3)	Applied DUT digital code-word	Applied reference DAC digital code-word	Measurement sense lines reference pin no. 25			Equations	Limits		Unit
			28 V_{CC}(V)	8(-----)1 MSB(-)LSB	12(---------)23 MSB(--------)LSB	Pin	Value	Unit		Min	Max	
3 T_A = -55°C	I_{CC}	075	+15.0	1111 1111	1111 1111 0000	28	I055	mA	I_{CC} = I055	+1	+25	mA
	I_{IL1}	076	"	0000 0000	0000 0000 0000	1	I056	µA	I_{IL1} = I056	-100	+100	µA
	I_{IL2}	077	"	" "	" " "	2	I057	"	I_{IL2} = I057	"	"	"
	I_{IL3}	078	"	" "	" " "	3	I058	"	I_{IL3} = I058	"	"	"
	I_{IL4}	079	"	" "	" " "	4	I059	"	I_{IL4} = I059	"	"	"
	I_{IL5}	080	"	" "	" " "	5	I060	"	I_{IL5} = I060	"	"	"
	I_{IL6}	081	"	" "	" " "	6	I061	"	I_{IL6} = I061	"	"	"
	I_{IL7}	082	"	" "	" " "	7	I062	"	I_{IL7} = I062	"	"	"
	I_{IL8}	083	"	" "	" " "	8	I063	"	I_{IL8} = I063	"	"	"
	I_{IH1}	084	"	1111 1111	1111 1111 0000	1	I064	"	I_{IH1} = I064 V_{IN}(LOGIC) = 2.4 V	"	"	"
	I_{IH2}	085	"	" "	" " "	2	I065	"	I_{IH2} = I065	"	"	"
	I_{IH3}	086	"	" "	" " "	3	I066	"	I_{IH3} = I066	"	"	"
	I_{IH4}	087	"	" "	" " "	4	I067	"	I_{IH4} = I067	"	"	"
	I_{IH5}	088	"	" "	" " "	5	I068	"	I_{IH5} = I068	"	"	"
	I_{IH6}	089	"	" "	" " "	6	I069	"	I_{IH6} = I069	"	"	"
	I_{IH7} , I_{IH8}	090 091	"	" "	" " "	7 8	I070 I071	"	I_{IH7} = I070 I_{IH8} = I071	"	"	"
	I_{IH9}	092	"	" "	" " "	1	I072	"	I_{IH13} = I072 V_{IN}(LOGIC) = 5.5 V	"	"	"
	I_{IH10}	093	"	" "	" " "	2	I073	"	I_{IH14} = I073	"	"	"
	I_{IH11}	094	"	" "	" " "	3	I074	"	I_{IH15} = I074	"	"	"
	I_{IH12}	095	"	" "	" " "	4	I075	"	I_{IH16} = I075	"	"	"
	I_{IH13}	096	"	" "	" " "	5	I076	"	I_{IH17} = I076	"	"	"
	I_{IH14}	097	"	" "	" " "	6	I077	"	I_{IH18} = I077	"	"	"
	I_{IH15}	098	"	" "	" " "	7	I078	"	I_{IH19} = I078	"	"	"
	I_{IH16}	099	"	" "	" " "	8	I079	"	I_{IH20} = I079	"	"	"
	I_{OSC}	100	"	" "	" " "	11	I080	mA	I_{OSC} = I080 V_{FSR} = 10.0 V		+30	mA
		101	+5.0	" "	" " "	11	I081	mA	I_{OSC} = I081 V_{FSR} = 2.56 V		+30	mA
	V_{ZE}	102	+15.0	0000 0000	0000 0000 0000	29	E021	V	V_{ZE1} = -.256*(E021) V_{FSR} = 10.0 V	-1.0	+1.0	LSB
		103	+5.0	0000 0000	0000 0000 0000	"	E022	"	V_{ZE2} = -1.00*(E022) V_{FSR} = 2.56 V	-1.0	+1.0	"
	V_{FS}	104	+15.0	1111 1111	1111 1111 0000	"	E023	"	V_{FS1} = -.256*(E023) V_{FSR} = 10.0 V	-1.75	+1.75	"
		105	+5.0	" "	" " "	"	E024	"	V_{FS2} = -1.00*(E024) V_{FSR} = 2.56 V	-1.75	+1.75	"
	$^+P_{SS1}$	106	+16.5	" "	" " "	"	E025	"				
		107	+14.0	" "	" " "	"	E026	"	$^+P_{SS1a}$ = .014*(E025-E026)	-.03	+.03	LSB/%PS
		108	+16.5	" "	" " "	"	E027	"	$^+P_{SS1b}$ = .014*(E026-E027)	"	"	"
	$^+P_{SS2}$	109	+5.5	" "	" " "	"	E028	"				
		110	+5.0	" "	" " "	"	E029	"	$^+P_{SS2a}$ = .10*(E028-E029)	"	"	"
		111	+4.5	" "	" " "	"	E030	"	$^+P_{SS2b}$ = .10*(E029-E030)	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltage (Fig 3)	Applied DUT digital code-word	Applied reference DAC digital code-word	Measurement sense lines reference pin no. 25			Equations	Limits		Unit
			28 V_{CC}(V)	8(-----)1 MSB(-)LSB	12(----------)23 MSB(-------)LSB	Pin	Value	Unit		Min	Max	
4 T_A = +25°C									For end-point linearity error, $L_E(N) = .256*(E(N+012)-V(ideal))$ $V(ideal) = N1*S-E030$ N1 = applied DUT code-word S = (E031-E030)/255			
		112	+15.0	0000 0000	0000 0000 0000	29	E030	V				
		113	"	1111 1111	1111 1111 0000	"	E031	"				
	L_{E1}	114	"	0000 0001	0000 0001 0000	"	E032	"	$L_{E1} = .256*(E032-1*S-E030)$	-.25	+.25	LSB
	L_{E2}	115	"	0000 0010	0000 0010 0000	"	E033	"	$L_{E2} = .256*(E033-2*S-E030)$	"	"	"
	L_{E4}	116	"	0000 0100	0000 0100 0000	"	E034	"	$L_{E4} = .256*(E034-4*S-E030)$	"	"	"
	L_{E8}	117	"	0000 1000	0000 1000 0000	"	E035	"	$L_{E8} = .256*(E035-8*S-E030)$	"	"	"
	L_{E16}	118	"	0001 0000	0001 0000 0000	"	E036	"	$L_{E16} = .256*(E036-16*S-E030)$	"	"	"
	L_{E32}	119	"	0010 0000	0010 0000 0000	"	E037	"	$L_{E32} = .256*(E037-32*S-E030)$	"	"	"
	L_{E64}	120	"	0100 0000	0100 0000 0000	"	E038	"	$L_{E64} = .256*(E038-64*S-E030)$	"	"	"
	L_{E128}	121	"	1000 0000	1000 0000 0000	"	E039	"	$L_{E128} = .256*(E039-128*S-E030)$	"	"	"
			Determine the code-word N1 and the value E040 of the maximum positive linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMP}	122	+15.0	(--N1---)	(--N1---) 1111	29	E040	V	$L_{EMP} = .256*(E040-N1*S-E030)$	"	"	"
			Determine the code-word N1 and the value E041 of the maximum negative linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMN}	123	+15.0	(--N1---)	(--N1---) 1111	29	E041	V	$L_{EMN} = .256*(E041-N1*S-E030)$	"	"	"
									For end-point linearity error, $L_E(N) = .256*(E(N+012)-V(ideal))$ $V(ideal) = N1*S-E042$ N1 = applied DUT code-word S = (E043-E042)/255			
		124	+5.0	0000 0000	0000 0000 0000	29	E042	V				
		125	"	1111 1111	1111 1111 0000	"	E043	"				
	L_{E1}	126	"	0000 0001	0000 0001 0000	"	E044	"	$L_{E1} = .256*(E044-1*S-E042)$	-.25	+.25	LSB
	L_{E2}	127	"	0000 0010	0000 0010 0000	"	E045	"	$L_{E2} = .256*(E045-2*S-E042)$	"	"	"
	L_{E4}	128	"	0000 0100	0000 0100 0000	"	E046	"	$L_{E4} = .256*(E046-4*S-E042)$	"	"	"
	L_{E8}	129	"	0000 1000	0000 1000 0000	"	E047	"	$L_{E8} = .256*(E047-8*S-E042)$	"	"	"
	L_{E16}	130	"	0001 0000	0001 0000 0000	"	E048	"	$L_{E16} = .256*(E048-16*S-E042)$	"	"	"
	L_{E32}	131	"	0010 0000	0010 0000 0000	"	E049	"	$L_{E32} = .256*(E049-32*S-E042)$	"	"	"
	L_{E64}	132	"	0100 0000	0100 0000 0000	"	E	"	$L_{E64} = .256*(E050-64*S-E042)$	"	"	"
	L_{E128}	133	"	1000 0000	1000 0000 0000	"	1	"	$L_{E128} = .256*(E051-128*S-E042)$	"	"	"

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TABLE III. Group A inspectic ntinued.

Subgroup	Symbol	Test no.	Applied voltage (Fig 3)	Applied DUT digital code-word	Applied reference DAC digital code-word	Measurem... sense lines reference pin no. 25			Equations	Limits		Unit
						Pin	Value	Unit		Min	Max	
			28 V_{CC}(V)	0(-----)11 MSJ(-)LSB	12(---------)23 MSB(-------)LSB							
T_A = +25°C 4			Determine the code-word N1 and the value E052 of the maximum positive linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMP}	134	+5.0	(--N1---)	(--N1---) 1111	29	E052	V	L_{EMP} = .256*(E053-N1*S-E042)	-.25	+.25	LSB
			Determine the code-word N1 and the value E053 of the maximum negative linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMN}	135	+5.0	(--N1---)	(--N1---) 1111	29	E053	V	L_{EMN} = .256*(E054-N1*S-E042)	-.25	+.25	LSB
	t_{DH}	136	+15.0		0000 0000 0000	"			See figures 3 and 4 1/	+10		ns
	t_{DS}	137	"		1111 1111 1111	"			See figures 3 and 4 2/	+200		ns
T_A = +125°C 5									For end-point linearity error, $L_E(N)$ = .256*(E(N*026)-V(ideal)) V(ideal) = N1*S-E064 N1 = applied DUT code-word S = (E055-E054)/255			
		138	"	0000 0000	0000 0000 0000	"	E064	V				
		139	"	1111 1111	1111 1111 0000	"	E055	V				
			Determine the code-word N1 and the value E056 of the maximum positive linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMP}	140	+15.0	(--N1---)	(--N1---) 1111	29	E056	V	L_{EMP} = .256*(E056-N1*S-E054)	-.375	+.375	LSB
			Determine the code-word N1 and the value E057 of the maximum negative linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMN}	141	+15.0	(--N1---)	(--N1---) 1111	29	E057	V	L_{EMN} = .256*(E057-N1*S-E054)	-.375	+.375	LSB
									For end-point linearity error, $L_E(N)$ = .256*(E(N*026)-V(ideal)) V(ideal) = N1*S-E058 N1 = applied DUT code-word S = (E059-E058)/255			
		142	+5.0	0000 0000	0000 0000 0000	"	E058	V				
		143	"	1111 1111	1111 1111 0000	"	E059	V				
			Determine the code-word N1 and the value E060 of the maximum positive linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMP}	144	+5.0	(--N1---)	(--N1---) 1111	29	E060	V	L_{EMP} = .256*(E060-N1*S-E058)	-.375	+.375	LSB
			Determine the code-word N1 and the value E061 of the maximum negative linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111).									
	L_{EMN}	145	+5.0	(--N1---)	(--N1---) 1111	29	E061	V	L_{EMN} = .256*(E061-N1*S-E058)	-.375	+.375	LSB
	t_{DH}	146	+15.0		0000 0000 0000	"			See figures 3 and 4 1/	+10		ns

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltage (Fig 3)	Applied DUT digital code-word	Applied reference DAC digital code-word	Measurement sense lines reference pin no. 25			Equations	Limits		Unit
			28 V_{CC}(V)	8(-----)1 MSB(-)LSB	12(---------)23 MSB(-------)LSB	Pin	Value	Unit		Min	Max	
6 T_A = -55°C									For end-point linearity error, $L_E(N) = .256*(E(N+140)-V(ideal))$ $V(ideal) = N1*S-E062$ N1 = applied DUT code-word $S = (E063-E062)/255$			
		148	+15.0	0000 0000	0000 0000 0000	29	E062	V				
		149	+15.0	1111 1111	1111 1111 0000	29	E063	V				
	L_{EMP}	150	+15.0	(--N1---)	(--N1---) 1111	29	E064	V	Determine the code-word N1 and the value E064 of the maximum positive linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111). $L_{EMP} = .256*(E064-N1*S-E062)$	-.375	+.375	LSB
	L_{EMN}	151	+15.0	(--N1---)	(--N1---) 1111	29	E065	V	Determine the code-word N1 and the value E065 of the maximum negative linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111). $L_{EMN} = .256*(E065-N1*S-E062)$	-.375	+.375	LSB
									For end-point linearity error, $L_E(N) = .256*(E(N+040)-V(ideal))$ $V(ideal) = N1*S-E066$ N1 = applied DUT code-word $S = (E067-E066)/255$			
		152	+5.0	0000 0000	0000 0000 0000	"	E066	"				
		153	+5.0	1111 1111	1111 1111 0000	"	E067	"				
	L_{EMP}	154	+5.0	(--N1---)	(--N1---) 1111	29	E068	V	Determine the code-word N1 and the value E068 of the maximum positive linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111). $L_{EMP} = .256*(E068-N1*S-E066)$	-.375	+.375	LSB
	L_{EMN}	155	+5.0	(--N1---)	(--N1---) 1111	29	E069	V	Determine the code-word N1 and the value E069 of the maximum negative linearity error for all combinations of the digital input code-words from (0000 0000) to (1111 1111). $L_{EMN} = .256*(E069-N1*S-E066)$	-.375	+.375	LSB
	t_{DH}	155	+15.0		0000 0000 0000	"			See figures 3 and 4 1/	+10		ns
	t_{DS}	157	+15.0		1111 1111 1111	"			See figures 3 and 4 3/	+300		ns

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltage	Applied DUT digital code-word	Applied reference DAC digital code-word	Measurement sense lines reference pin no. 21			Equations	Limits		Unit
			28 V_{CC}(V)	8(-----)1 MSB(-)LSB	12(---------)23 MSB(-------)LSB	Pin	Value	Unit		Min	Max	
9 T_A = +25°C	t_{SLH1}	158	+15.0				t001		t_{SLH1} = t001 Figures 4 and 5		+3.0	µs
	t_{SHL1}	159	+15.0				t002		t_{SHL1} = t002 Figures 4 and 5		+3.0	µs
	t_{SLH2}	160	+5.0				t003		t_{SLH2} = t003 Figures 4 and 5		+1.5	µs
	t_{SHL2}	161	+5.0				t004		t_{SHL2} = t004 Figures 4 and 5		+1.5	µs
12 T_A = +25°C	N_0	162	+15.0	1111 1111			E070		N0 = E070 Figure 6		+250	µV rms

1/ a. Set $\overline{CS}$ and $\overline{CE}$ to digital "1".
b. Set up data converter for full scale voltage output.
c. Set input code = 0000 0000.
d. Strobe $\overline{CS}$ and $\overline{CE}$ low at t = 0 ns, and high at t = 200 ns.
e. Strobe input code to 1111 1111 at t = 160 ns.
f. Offset voltage error (±0.5 LSB).

2/ a. Set $\overline{CS}$ and $\overline{CE}$ to digital "0".
b. Set up data converter for 0 volt output.
c. At t = 0 ns, strobe input code to 1111 1111.
d. At t = 200 ns, strobe $\overline{CS}$ and $\overline{CE}$ to digital "1".
e. Full scale voltage error (±1.0 LSB).

3/ a. Set $\overline{CS}$ and $\overline{CE}$ to digital "0".
b. Set up data converter for 0 volt output.
c. At t = 0 ns, strobe input code to 1111 1111.
d. At t = 300 ns, strobe $\overline{CS}$ and $\overline{CE}$ to digital "1".
e. Full scale voltage error (±1.0 LSB).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) Class S devices: Dynamic tests (test condition D) using the circuit shown on figure 7; test duration shall be 240 hours minimum.

(2) Class B devices: Test condition D, using the circuit shown on figure 7, or test condition C, using the circuit shown on figure 8, or test condition F, using the circuit shown on figure 7.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 7, 8, 10, and 11 shall be omitted.

c. A special subgroup shall be added to group A inspection for all classes and it shall consist of the conditions and limits as specified in table III. Subgroup 12, LTPD shall be 15.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 7. If the alternate burn-in conditions are used, the circuit shown on figure 8 shall be used.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class B devices (method 1005 of MIL-STD-883). Test condition D, using the circuit shown on figure 8, or test condition F using the circuit shown on figure 7, or test condition C using the circuit shown on figure 8.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883.

~~End-point electrical parameters shall be as specified in table II herein.~~

TABLE IV. Group C end-point electrical parameters, T_A = +25°C.

Table III test number	Symbol	Delta 1/	End-point limits		Units
			Min	Max	
28	V_{ZE}	0.1	-0.6	0.6	LSB
30, 31	V_{FS}	0.75	-1.75	1.75	LSB
114-121	L_E 2/	0.125	-0.375	0.375	LSB
126-133	L_E 2/	0.125	-0.375	0.375	LSB

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).
2/ End-point linearity error tests shall be run on the following codes:

Test number	Code number
114, 126	0000 0001
115, 127	0000 0010
116, 128	0000 0100
117, 129	0000 1000
118, 130	0001 0000
119, 131	0010 0000
120, 132	0100 0000
121, 133	1000 0000

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical tests circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the ground terminal of the device under test DUT. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test and burn-in cooldown procedure. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within +10°C of their power stable condition at room temperature prior to removal of the bias.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

D_{LE}	Differential linearity error
I_{CC}	Input current for voltage supply
I_{IL}	Digital low level input current
I_{IH}	Digital high level input current
I_{OSC}	Output short circuit current
L_E	Bit linearity error
LSB	Least significant bit
MSB	Most significant bit
N_O	Output noise voltage
O_{JC}	Junction-to-case thermal resistance
O_{JA}	Junction-to-ambient thermal resistance
P_{SS}	Power supply sensitive
T_A	Ambient temperature
t_{DH}	Data hold time
t_{DS}	Data setup time
t_{SHL}	High-to-low output settling time
t_{SLH}	Low-to-high output settling time
V_{FS}	Full scale voltage
V_{FSR}	Full scale voltage range
V_{IN}	Digital input voltage
V_{OS}	Offset voltage
Z_{REF}	Reference input resistance

Differential linearity error. Differential linearity error is the difference between actual separation of two adjacent output levels and the ideal separation of one LSB.

Full-scale output voltage range. The full-scale output voltage range is the output voltage change between the input codes of all zeros to ones plus the voltage equivalent to one additonal LSB.

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Least significant bit. The LSB of the digital code-word is the bit that causes the least change to the output level. When the LSB changes state, the ideal output level changes by the value of the full scale range divided by 256.

Monolithic. Integrated circuit monolithic technology includes all of the components; such as transistors, resistors, capacitor, etc., on a single silicon substrate in a DIP.

Monotonicity. An ADC or DAC transfer function is monotonic if the ratio of the incremental change in the output to an incremental change in the input does not change sign over the entire full scale range.

Most significant bit. The MSB of the digital codeword is the bit that causes the most change to the output level. When the MSB changes state, the output level changes by one-half of the full scale output range.

Offset. Offset is the output voltage for an input code of all zeros.

Power supply sensitivity at full scale. Power supply sensitivity at full scale is the change in the full scale output level with a change in the power supply voltage.

Resolution. Resolution of a DAC is the smallest ideal increment between two adjacent output levels over the full-scale range of the converter.

Settling time. Settling time is the elapsed time from the application of a full-scale step input to the DAC to the time when the output has entered and remained within a 1/2 LSB error band around its final value.

Unipolar mode. Unipolar mode is the D/A converter operation mode that includes output voltages or currents of one polarity only.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	AD558T

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC
Air Force - 80

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0950)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/144	MICROCIRCUITS, LINEAR, MICROPROCESSOR COMPATIBLE, 8-BIT DIGITAL-TO-ANALOG CONVERTER, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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